Question 21 (10 marks) Use the Question 21 writing booklet.

(a) What is *data redundancy* in a database? Include an example in your answer. **2**

(b) A database is to be developed to manage a graduation dinner. Students will be able to invite guests to the dinner.

The data dictionary for the booking table is shown. Some information about the fields is missing.

Field name	*Data type*	*Field size*	*Description*
Student_ID	Number	5	
First_Name	Text	25	
Last_Name	Text	25	
Phone_Number			Student's mobile number eg 0499123456
No_of_guests	Number	1	
Deposit_Paid			Shows whether a deposit has been paid

(i) Copy the table below into your writing booklet. **3**

In your table, recommend an appropriate data type and field size for Phone_Number and Deposit_Paid, and justify each choice of data type.

Field name	*Data type*	*Field size*	*Justification for choice of data type*
Phone_Number			
Deposit_Paid			

(ii) If there are 200 records, show how the size of the booking table (in megabytes) can be calculated. **2**

(iii) Write a SQL statement to list the names in alphabetical order (A–Z) by Last_Name of the students who have more than one guest. Use the following keywords in your answer. **3**

SELECT

FROM

WHERE

ORDER BY